Zhi Su Undergraduate

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Education

Institute for Interdisciplinary Information Sciences, Tsinghua University Yao Class
Undergraduate Student in Computer Science and Technology 2022 - 2026

- GPA: 3.9/4.0, Rank: 9/94.

Publications

1. **Zhi Su**, Bike Zhang, Nima Rahmanian, Yuman Gao, Qiayuan Liao, Caitlin Regan, Koushil Sreenath, S. Shankar Sastry. "HITTER: A HumanoId Table TEnnis Robot via Hierarchical Planning and Learning". *ArXiv*, 2025.
2. **Zhi Su***, Yuman Gao*, Emily Lukas*, Yunfei Li, Jiaze Cai, Faris Tulbah, Fei Gao, Chao Yu, Zhongyu Li, Yi Wu, Koushil Sreenath. "Toward Real-World Cooperative and Competitive Soccer with Quadrupedal Robot Teams". *CoRL*, 2025.
3. **Zhi Su***, Xiaoyu Huang*, Daniel Ordoñez-Apraez, Yunfei Li, Zhongyu Li, Qiayuan Liao, Giulio Turrisi, Massimiliano Pontil, Claudio Semini, Yi Wu, Koushil Sreenath. "Leveraging Symmetry in RL-based Legged Locomotion Control." *IROS*, 2024.
4. Yunfei Li, Xiao Ma, Jiafeng Xu, Yu Cui, Zhongren Cui, Zhigang Han, Liqun Huang, Tao Kong, Yuxiao Liu, Hao Niu, Wanli Peng, Jingchao Qiao, Zeyu Ren, Haixin Shi, **Zhi Su**, Jiawen Tian, Yuyang Xiao, Shenyu Zhang, Liwei Zheng, Hang Li, Yonghui Wu. "GR-RL: Going Dexterous and Precise for Long-Horizon Robotic Manipulation." *ArXiv*, 2025.

Research Experience

Tsinghua University | Beijing, China 2023.09 - 2026.06

- Advisor: Prof. Yi Wu.

University of California, Berkeley | California, USA 2025.03 - 2025.08

- Advisor: Prof. Koushil Sreenath and Prof. Shankar Sastry.

Internship

ByteDance Seed Robotics | Beijing, China 2025.10 - 2026.06

- Working on the next generation of VLA models and real-world RL.

Projects

Utilization and Effects of Human Data in Training Vision–Language–Action Models
Bachelor's Thesis, Tsinghua University 2025 Fall – 2026 Spring

- Developed a Qwen3.5-based VLA that aligns egocentric human hand trajectories with robot end-effector actions, and studied human–robot co-training through visual augmentation, embodiment-specific projections, and mechanistic probing.

Score-Matching Motion Priors for Unitree G1
GitHub Repository 2026 Spring

- Reproduced Score-Matching Motion Priors on Unitree G1 by pretraining diffusion motion priors and reusing their frozen scores as PPO guidance for locomotion, steering, navigation, and get-up tasks.

Privileged Observations for RGB-Based Robotic Manipulation
GitHub Repository 2024 Fall

- Developed an asymmetric PPO pipeline that distills privileged state observations into RGB features for robust robotic manipulation and evaluated it on custom ManiSkill tasks.

Projects (continued)

Rigid-Body Coupling in MLS-MPM Simulation

Course Project for Advanced Computer Graphics 2024 Fall

- Extended an MLS-MPM simulator with rigid-body dynamics and two-way coupling between rigid objects and continuum materials.

CyberDog 2 Quadrupedal and Bipedal Dribbling

Course Project for Deep Reinforcement Learning 2024 Spring

- Trained deep reinforcement learning policies for quadrupedal and bipedal locomotion and ball dribbling with the CyberDog 2 platform in simulation.

LLM-Based Situation Puzzles

GitHub Repository 2024 Spring

- Built a multimodal situation-puzzle game with LLM host and player agents, AI-vs-AI interaction, tool-generated image hints, and speech output.

LLM-Based Music Recommender

GitHub Repository 2023 Fall

- Built a conversational music recommender combining ChatGLM3-6B, a local RAG knowledge base, web search, and NetEase Cloud Music APIs.

Awards and Honors

- **Academic Scholarship**, Tsinghua University 2023.10
- **Academic Scholarship**, Tsinghua University 2024.10
- **Yao Award (recognition prize)**, Tsinghua University 2025.09
- **Academic Scholarship**, Tsinghua University 2025.10

Technical Strengths

Advanced: Pytorch, IsaacGym, IsaacLab
Intermediate: Git, LaTeX, Linux
Basic: C, C++, Python

Research Interest

Deep Reinforcement Learning
Quadrupedal Robots
Humanoid Robots
Embodied AI